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Hydrogen production from gasification of sour gas

Abstract

A sour gas stream is sub-stoichiometrically combusted to produce soot and a sour syngas stream. At least 10% of the carbon in the sour gas stream is converted into the soot. At least a portion of the hydrogen sulfide of the sour syngas stream is reacted with sulfur dioxide to produce a syngas stream comprising the carbon dioxide, the carbon monoxide, the hydrogen, water, elemental sulfur vapor, a residual portion of the hydrogen sulfide, and a residual portion of the sulfur dioxide. The syngas stream is reacted with steam to produce a shifted sour gas stream including more carbon dioxide, more hydrogen, more hydrogen sulfide, and less carbon monoxide in comparison to the syngas stream. Water and hydrogen sulfide is separated from the shifted sour gas stream to produce a sweet gas stream. The sweet gas stream is separated into a hydrogen product stream and an exhaust stream.

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Background/Summary

TECHNICAL FIELD

(1) This disclosure relates to gasification, and in particular, gasification of sour gas to produce hydrogen.

BACKGROUND

(2) Hydrogen sulfide and carbon dioxide exist in various gas streams, including natural gas. Oil or gas that contains significant amounts of sulfur compounds like hydrogen sulfide is considered “sour”, and oil refineries and gas processing plants utilize “sweetening” processes to remove such sulfur compounds. Typical sulfur recovery processes selectively remove acid gases, such as hydrogen sulfide and carbon dioxide, from gas mixtures. The hydrogen sulfide can then be converted into elemental sulfur and, in some cases, sulfur dioxide. The carbon dioxide, on the other hand, is typically released into the atmosphere without further use.

SUMMARY

(3) This disclosure describes technologies relating to gasification of sour gas to produce hydrogen. Certain aspects of the subject matter described can be implemented as a method. A sour gas stream is combusted within a reactor in the presence of oxygen to produce soot and a sour syngas stream comprising carbon dioxide, carbon monoxide, hydrogen, and hydrogen sulfide. The sour gas stream includes hydrogen sulfide and at least one hydrocarbon. Combusting the sour gas stream within the reactor includes sub-stoichiometric combustion of the sour gas stream, wherein the sour gas stream has a molar content of carbon (C) and a molar content of hydrogen (H), wherein a molar content of oxygen gas in the reactor is in a range of from 10% to 70% of $C+H/4$. The sub-stoichiometric combustion of the sour gas stream within the reactor converts at least 10% of the carbon in the sour gas stream into the soot. At least a portion of the hydrogen sulfide of the sour syngas stream is reacted with sulfur dioxide to produce a syngas stream that includes the carbon dioxide, the carbon monoxide, the hydrogen, water, elemental sulfur vapor, a residual portion of the hydrogen sulfide, and a residual portion of the sulfur dioxide. Reacting at least the portion of the hydrogen sulfide of the sour syngas stream with sulfur dioxide produces a sulfur stream comprising elemental sulfur liquid. The syngas stream is contacted, in the presence of steam, with a water-gas shift hydrogenation catalyst in a shift reactor to convert at least a portion of the carbon monoxide and steam to carbon dioxide and hydrogen to produce a shifted sour gas stream. Contacting the syngas stream with the water-gas shift hydrogenation catalyst in the shift reactor in the presence of steam reduces the elemental sulfur vapor and the residual portion of the sulfur dioxide into hydrogen sulfide. The shifted sour gas stream is substantially free of elemental sulfur and sulfur dioxide. Water is separated from the shifted sour gas stream to produce a dehydrated sour gas stream. The dehydrated sour gas stream is contacted with a lean amine solvent to extract the residual portion of the hydrogen sulfide from the dehydrated sour gas stream into the lean amine solvent to produce a sweet gas stream and a rich amine solvent. The sweet gas stream includes the carbon dioxide and the hydrogen. The sweet gas stream can be saturated with water (steam) at the operating conditions of the sweet gas stream. For example, the sweet gas stream can include about 3-5 volume percent steam. The residual portion of the hydrogen sulfide is boiled off from the rich amine solvent to

produce an acid gas stream and regenerate the lean amine solvent. The acid gas stream includes the residual portion of the hydrogen sulfide. The sweet gas stream is separated to produce a hydrogen product stream and an exhaust stream. The hydrogen product stream includes a majority of the hydrogen from the sweet gas stream, and the exhaust stream includes a remainder of the sweet gas stream.

(4) This, and other aspects, can include one or more of the following features. In some implementations, at least a portion of heat generated from sub-stoichiometric combustion of the sour gas stream within the reactor is directed to a boiler to generate steam and cool the soot and sour syngas to a temperature in a range of from about 250 degrees Celsius ($^{\circ}\text{C}$.) to about 500°C . In some implementations, at least a portion of the steam stream is sourced from the steam generated by the boiler. In some implementations, at least a portion of the steam generated by the boiler is flowed through a steam turbine, and the steam turbine generates electrical power in response to flow of the portion of the steam through the turbine. In some implementations, separating water from the shifted sour gas stream includes contacting the shifted sour gas stream with a water stream within a quench tower and discharging a reject water stream and the dehydrated sour gas stream from the quench tower. In some implementations, at least a portion of the acid gas stream is recycled to the reactor. In some implementations, at least a portion of the exhaust stream is recycled to the reactor. In some implementations, a second portion of the exhaust stream is sequestered within a subterranean formation to avoid releasing carbon dioxide to the atmosphere. In some implementations, a high pressure raw sour gas stream is processed in a feed pre-treatment unit. Processing the high pressure raw sour gas stream in the feed pre-treatment unit can include separating water from the high pressure raw sour gas stream and condensing natural gas liquids from the high pressure raw sour gas stream to produce a natural gas liquids (NGL) stream and the sour gas stream. In some implementations, the sour gas stream is flowed through a turboexpander prior to combusting the sour gas stream within the reactor. The turboexpander can generate electrical power in response to flow of the sour gas stream through the turboexpander.

(5) Certain aspects of the subject matter described can be implemented as a system. The system includes a sour gas stream, an oxidizing stream, a combustion reactor, a mechanical separator, a Claus unit, a sour shift/hydrogenation reactor, a quench tower, an amine unit, and a purifier. The sour gas stream includes hydrogen sulfide and at least one hydrocarbon. The sour gas stream has a molar flow rate of carbon (C) and a molar flow rate of hydrogen (H). The oxidizing stream includes oxygen gas and has a molar flow rate of oxygen gas (O_{2}) in a range of from 10% to 70% of $\text{C}+\text{H}/4$. The combustion reactor is configured to receive the sour gas stream and the oxidizing stream. The combustion reactor is configured to combust the sour gas stream to produce soot and sour syngas. Sub-stoichiometric combustion of the sour gas stream within the combustion reactor converts at least 10% of the carbon in the sour gas stream into soot. The mechanical separator is configured to separate the soot from the sour syngas to produce a sour syngas stream comprising carbon dioxide, carbon monoxide, hydrogen, and hydrogen sulfide. The Claus unit is configured to receive the sour syngas stream and a sulfur dioxide stream that includes sulfur dioxide. The Claus unit is configured to react at least a portion of the hydrogen sulfide of the sour syngas stream with the sulfur dioxide of the sulfur dioxide stream to produce elemental sulfur and water. The Claus unit is configured to discharge a sulfur stream that includes the elemental sulfur. The Claus unit is configured to discharge a syngas stream that includes the carbon dioxide, the carbon monoxide, the hydrogen, the water, elemental sulfur vapor, a residual portion of the hydrogen sulfide, and a residual portion of the sulfur dioxide. The sour shift/hydrogenation reactor is configured to receive the syngas stream and steam. The sour shift/hydrogenation reactor includes a water-gas shift hydrogenation catalyst. The water-gas shift hydrogenation catalyst is configured to convert at least a portion of the carbon monoxide of the syngas stream and steam to carbon dioxide and hydrogen to produce a shifted sour gas stream. The water-gas shift hydrogenation catalyst is configured to reduce the elemental sulfur vapor and the residual portion of the sulfur dioxide into hydrogen

sulfide. The shifted sour gas stream is substantially free of elemental sulfur and sulfur dioxide. The quench tower is configured to receive the shifted sour gas stream and separate water from the shifted sour gas stream to produce a dehydrated sour gas stream. The amine unit is configured to receive the dehydrated sour gas stream. The amine unit is configured to contact the dehydrated sour gas stream with a lean amine solvent to extract the residual portion of the hydrogen sulfide from the dehydrated sour gas stream into the lean amine solvent to produce a sweet gas stream and a rich amine solvent. The sweet gas stream includes the carbon dioxide and the hydrogen. The amine unit is configured to boil off the residual portion of the hydrogen sulfide from the rich amine solvent to produce an acid gas stream and regenerate the lean amine solvent. The acid gas stream includes the residual portion of the hydrogen sulfide. The purifier is configured to receive the sweet gas stream and separate hydrogen from the sweet gas stream to produce a hydrogen product stream and an exhaust stream. The hydrogen product stream includes a majority of the hydrogen from the sweet gas stream. The exhaust stream includes a remainder of the sweet gas stream. The combustion reactor is configured to receive at least a portion of the exhaust stream.

(6) This, and other aspects, can include one or more of the following features. In some implementations, the system includes a boiler configured to use heat generated from combustion of the sour gas stream within the combustion reactor to generate steam and cool the soot and sour syngas to a temperature in a range of from about 250 degrees Celsius ($^{\circ}\text{C}$.) to about 500°C . In some implementations, the sour shift/hydrogenation reactor is configured to receive at least a portion of the steam generated by the boiler. In some implementations, the system includes a steam turbine configured to receive at least a portion of the steam generated by the boiler. The steam turbine can be configured to generate electrical power in response to the portion of the steam flowing through the steam turbine. In some implementations, the combustion reactor is configured to receive at least a portion of the acid gas stream. In some implementations, a second portion of the exhaust stream is sequestered within a subterranean formation to avoid releasing carbon dioxide to the atmosphere. In some implementations, the system includes a feed pre-treatment unit upstream of the combustion reactor. The feed pre-treatment unit can be configured to process a high pressure raw sour gas stream to separate water and natural gas liquids from the high pressure raw sour gas stream to produce a reject water stream, a natural gas liquids (NGL) stream, and the sour gas stream. In some implementations, the system includes a turboexpander downstream of the feed pre-treatment unit and upstream of the combustion reactor. The turboexpander can be configured to receive the sour gas stream and generate electrical power in response to flow of the sour gas stream through the turboexpander. The turboexpander can be configured to discharge the sour gas stream to the combustion reactor.

(7) Certain aspects of the subject matter described can be implemented as a method. A sour gas stream is combusted in the presence of oxygen to produce soot and a sour syngas stream comprising carbon dioxide, carbon monoxide, hydrogen, and hydrogen sulfide. The sour gas stream includes hydrogen sulfide and at least one hydrocarbon. Combusting the sour gas stream includes sub-stoichiometric combustion of the sour gas stream. The sour gas stream has a molar content of carbon (C) and a molar content of hydrogen (H). A molar content of the oxygen that is present is in a range of from 10% to 70% of $\text{C}+\text{H}/4$. The sub-stoichiometric combustion of the sour gas stream converts at least 10% of the carbon in the sour gas stream into the soot. At least a portion of the hydrogen sulfide of the sour syngas stream is reacted with sulfur dioxide to produce a syngas stream that includes the carbon dioxide, the carbon monoxide, the hydrogen, water, elemental sulfur vapor, a residual portion of the hydrogen sulfide, and a residual portion of the sulfur dioxide. Reacting at least the portion of the hydrogen sulfide of the sour syngas stream with sulfur dioxide produces a sulfur stream comprising elemental sulfur liquid. At least a portion of the carbon monoxide of the syngas stream is reacted with steam in the presence of a water-gas shift hydrogenation catalyst to produce a shifted sour gas stream. The shifted gas sour stream includes more carbon dioxide, more hydrogen, more hydrogen sulfide, and less carbon monoxide in

comparison to the syngas stream prior to reacting the carbon monoxide. The shifted sour gas stream is substantially free of elemental sulfur and sulfur dioxide. Water and the residual portion of the hydrogen sulfide are separated from the shifted sour gas stream to produce a sweet gas stream. The sweet gas stream is separated into a hydrogen product stream and an exhaust stream. The hydrogen product stream includes a majority of the hydrogen from the sweet gas stream. The exhaust stream includes a remainder of the sweet gas stream.

(8) This, and other aspects, can include one or more of the following features. In some implementations, at least a portion of heat generated from combustion of the sour gas stream is directed to a boiler to generate steam and using at least a portion of the generated steam to react with at least the portion of the carbon monoxide of the syngas stream in the presence of the water-gas shift hydrogenation catalyst.

(9) The details of one or more implementations of the subject matter of this disclosure are set forth in the accompanying drawings and the description. Other features, aspects, and advantages of the subject matter will become apparent from the description, the drawings, and the claims.

Description

DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a schematic diagram of an example system for producing hydrogen from gasification of sour gas.

(2) FIG. 2 is a flowchart of an example method for producing hydrogen from gasification of sour gas.

(3) FIG. 3 is a flowchart of an example method for producing hydrogen from gasification of sour gas.

DETAILED DESCRIPTION

(4) This disclosure describes hydrogen production that includes sub-stoichiometric combustion of sour gas (sour gas gasification), a multi-stage Claus process, a hydrogenation reactor for water-gas shift and sulfur compound reduction and hydrolysis, and a low pressure amine system. The sub-stoichiometric combustion of hydrocarbons produces sour syngas (carbon monoxide, hydrogen, and possibly sulfur dioxide (SO_2) and sulfur) and maximizes soot production for the purpose of reducing the amount of carbon dioxide produced in the combustion process. The heat from combustion is used to generate steam. Unreacted hydrogen sulfide is converted and recovered as elemental sulfur in the multi-stage Claus process. The syngas is then processed in a hydrogenation reactor to convert the carbon monoxide into additional hydrogen via the water-gas shift reaction and to reduce and hydrolyze sulfur compounds back to H_2S . Steam (produced by the heat from combustion) may be added to the hydrogenation reactor to promote conversion of carbon monoxide into hydrogen. The outlet stream from the hydrogenation reactor is quenched to cool the outlet stream and to separate water. The outlet stream is then further processed in a low pressure absorber for the removal of H_2S . The remaining portion of the outlet stream (excluding water) is a hydrogen-rich stream. The hydrogen-rich stream can be further processed (for example, by a membrane or a pressure swing adsorption process) to separate contaminants and purify the hydrogen-rich stream. Residual carbon monoxide, carbon dioxide, light hydrocarbons (such as methane), and hydrogen sulfide can be recycled for re-processing and increasing the overall efficiency of the system.

(5) The subject matter described in this disclosure can be implemented in particular implementations, so as to realize one or more of the following advantages. The processes and systems described can minimize or eliminate carbon dioxide production when converting hydrocarbons into hydrogen. For example, these processes and systems implement partial oxidation (sub-stoichiometric combustion) of hydrocarbons to specifically target maximum soot production

and minimum carbon dioxide production. The soot that is produced can be processed into other useful forms of carbon and/or sold to the market. These processes and systems can implement process and heat integration, such that heat produced within the process/system can be used elsewhere in the process/system. As such, these processes and systems can be less energy intensive in comparison to conventional hydrogen production processes. Carbon dioxide produced by the system can be recycled within the system and/or sequestered in a subterranean formation to avoid release of greenhouse gases into the atmosphere. These systems and methods not only produce hydrogen, which is a useful gas, but also sweeten sour gas (that is, removes sulfur compounds such as H_2S from the gas stream) and produce electrical power, which can be used internally by the respective systems and methods, be provided to another end user, be provided to a power grid, or any combinations of these.

(6) FIG. 1 depicts an example system **100** for hydrogen production from gasification of sour gas. The system **100** can be carbon neutral in that carbon dioxide emissions from the system **100** are at or close to net zero. That is, all or substantially all of the carbon dioxide produced by the system **100** is subsequently used by the system **100** and/or sequestered in a subterranean formation, such that no (or a negligible amount of) carbon dioxide is released to the atmosphere. The system **100** includes a combustion reactor **110**, a mechanical separator **120**, a shift/hydrogenation reactor **130**, a quench tower **140**, a purifier **150**, a Claus unit **160**, and an amine unit **170**. The system **100** includes a sour gas stream **101** and an oxidizing stream **103**. The sour gas stream **101** includes hydrogen sulfide (H_2S) and at least one hydrocarbon (for example, methane, ethane, propane, or butane). For example, the sour gas stream **101** is a natural gas stream including methane and H_2S . As another example, the sour gas stream **101** is a sour gas stream that has been separated from a liquefied petroleum gas (LPG) stream that includes propane, butane, or both. A molar flow rate of carbon in the sour gas stream **101** is C . A molar flow rate of hydrogen in the sour gas stream **101** is H . The oxidizing stream **103** includes oxygen gas (O_2). In some implementations, the oxidizing stream **103** is predominantly oxygen (for example, high purity oxygen with an oxygen concentration of greater than 99%). In some implementations, the oxidizing stream **103** is air, which includes oxygen. A molar flow rate of oxygen gas (O_2) is O . The combustion reactor **110** is configured to receive the sour gas stream **101** and the oxidizing stream **103** and is configured to combust the sour gas stream **101** in the presence of steam to produce soot and sour syngas. In some implementations, the combustion reactor **110** is configured to combust the sour gas stream **101** in the absence of steam to produce soot and sour syngas (that is, in some cases, steam is not necessary to produce the soot and sour syngas). In some implementations, the system **100** includes a first steam stream **105**, which flows to the combustion reactor **110** to provide temperature control and replenish steam supply for steam reforming reaction(s) within the combustion reactor **110**.

(7) The ratio of oxygen in the oxidizing stream **103** to hydrocarbons in the sour gas stream **101** is sub-stoichiometric in relation to combustion. Sub-stoichiometric combustion is also referred to as partial oxidation. O is in a range of from 10% to 70% of $C+H/4$. As one example, the sour gas stream **101** can include 100 moles per hour of methane, and then for stoichiometric combustion (complete 100% combustion), the balanced chemical reaction is:



(8) In other words, for stoichiometric combustion for a methane stream, so 2 moles of oxygen gas (O_2) is needed per mole of methane (CH_4) (which confirms: $C=1$; $H=4$; $C+H/4=2$). For the sub-stoichiometric combustion of the sour gas stream **101**, the rate of oxygen flow in the oxidizing stream **103** is from 10% to 70% of the oxygen rate necessary for stoichiometric combustion. For this specific example (where the sour gas stream **101** includes 100 moles per hour of methane), the oxidizing stream **103** includes from 20 moles per hour (10%) to 140 moles per hour (70%) of oxygen gas (O_2). Sub-stoichiometric combustion of the sour gas stream **101** produces additional products (apart from just carbon dioxide and water), such as hydrogen (H_2), carbon monoxide (CO), and solid carbon (C).

(9) Providing a sub-stoichiometric ratio of oxygen to hydrocarbons in the combustion reactor **110** increases the rate of soot production within the combustion reactor **110**, which in turn reduces the rate of carbon dioxide production within the combustion reactor **110**. The combustion reactor **110** is configured to convert at least 10% of the carbon in the sour gas stream **101** into soot (as opposed to carbon dioxide) as a result of the sub-stoichiometric combustion of the sour gas stream **101** within the combustion reactor **110**. In some implementations, the combustion reactor **110** is configured to convert at least 20%, at least 30%, at least 40%, or at least 50% of the carbon in the sour gas stream **101** into soot as a result of the sub-stoichiometric combustion of the sour gas stream **101** within the combustion reactor **110**. For example, the combustion reactor **110** is configured to convert from about 10% to about 80%, from about 20% to about 70%, from about 30% to about 60%, from about 40% to about 50% of the carbon in the sour gas stream **101** into soot as a result of the sub-stoichiometric combustion of the sour gas stream **101** within the combustion reactor **110**.

(10) The mechanical separator **120** receives the soot and the sour syngas that was produced in the combustion reactor **110**. The mechanical separator **120** is configured to separate the soot (solid carbon) from the sour syngas to produce a sour syngas stream **121**. The sour syngas stream **121** includes carbon dioxide, carbon monoxide, hydrogen, and hydrogen sulfide. In some cases, the sour syngas stream **121** includes a low level of sulfur dioxide (SO_2). In some implementations, the sour syngas stream **121** includes unreacted hydrocarbons originating from the sour gas stream **101**, such as methane and/or other hydrocarbon(s). The mechanical separator **120** can capitalize on the effects of gravity to separate the soot from the sour syngas by taking advantage of the difference in density of the components. The mechanical separator **120** can include, for example, a cyclonic separator. In some implementations, the mechanical separator **120** includes a filter (such as a bag filter) that allows gaseous components (such as the syngas) to pass through the filter while preventing solid components (such as the soot) from passing through the filter to separate the soot from the sour syngas. In some implementations, the mechanical separator **120** includes a water spray injection and/or a wash column that can facilitate coalescence of the soot particles to more easily separate the soot from the sour syngas. The soot **123** that is separated from the mechanical separator **120** can be disposed, used for power generation, or processed, for example, to produce useful forms of carbon, such as carbon black, synthetic graphite, carbon filaments/fiber, and/or carbon nanostructures (such as carbon nanotubes or carbon nanofibers). In some implementations, the system **100** includes a preheater (not shown) downstream of the mechanical separator **120** and upstream of the Claus unit **160**. In cases where the system **100** includes the preheater, the preheater can re-heat the sour syngas stream **121** to an operating temperature suitable for promoting the reaction(s) occurring in the Claus unit **160**. The preheater can, for example, utilize steam to re-heat the sour syngas stream **121**.

(11) The Claus unit **160** is configured to receive the sour syngas stream **121** and a sulfur dioxide stream **127** that includes sulfur dioxide. In some implementations, the sulfur dioxide stream **127** is provided at a flow rate that supplies sufficient sulfur dioxide to achieve a molar ratio of 2:1 for hydrogen sulfide to sulfur dioxide ($\text{H}_2\text{S}:\text{SO}_2$) entering the Claus unit **160**. In some implementations, the sour syngas stream **121** mixes with the sulfur dioxide stream **127** before entering the Claus unit **160**. In some implementations, the sour syngas stream **121** and the sulfur dioxide stream **127** separately enter the Claus unit **160** and mix within the Claus unit **160**. The Claus unit **160** is configured to react at least a portion of the hydrogen sulfide of the sour syngas stream **121** with the sulfur dioxide of the sulfur dioxide stream **127** to produce sulfur and water. The Claus unit **160** is configured to discharge a sulfur stream **129** that includes the sulfur produced in the Claus unit **160**. The Claus unit **160** is configured to discharge a syngas stream **121'** that includes the carbon dioxide, the carbon monoxide, the hydrogen, the water, sulfur (vapor), and a residual portion of the hydrogen sulfide and sulfur dioxide that did not react in the Claus unit **160**. The syngas stream **121'** exiting the Claus unit **160** has a lower hydrogen sulfide content in comparison to the sour syngas stream **121** entering the Claus unit **160**. The sulfur produced by the

Claus unit **160** can be separated from the gas to produce the sulfur stream **129**. In some implementations, the Claus unit **160** converts at least 90% or at least 95% of the hydrogen sulfide present in the sour syngas stream **121** (entering the Claus unit **160**) into elemental sulfur. The syngas stream **121'** exiting the Claus unit **160** continues to the shift/hydrogenation reactor **130**.

(12) The Claus unit **160** includes a Claus catalyst **162** that accelerates the reaction between the hydrogen sulfide of the sour syngas stream **121** with the sulfur dioxide of the sulfur dioxide stream **127** to produce sulfur and water. The Claus catalyst **162** can include activated aluminum (III) or titanium (IV) oxide. In some implementations, the Claus unit **160** includes a single-stage Claus process that includes a single reactor train. The reactor train can include a preheater, a Claus reactor including the Claus catalyst **162**, and a condenser. The preheater heats the process stream (for example, the sour syngas stream **121**) to a temperature that facilitates the conversion of hydrogen sulfide into elemental sulfur and sulfur dioxide (and water) in the presence of the Claus catalyst **162**. In some implementations, the preheater heats the process stream (sour syngas stream **121**) to provide a catalytic converter maximum bottom bed temperature in a range of from about 300 degrees Celsius (° C.) to about 340° C. After preheating by the preheater, the process stream (sour syngas stream **121**) flows to the Claus reactor. The Claus catalyst **162** promotes the conversion of hydrogen sulfide into elemental sulfur and sulfur dioxide (and water). The process stream exiting the Claus reactor then flows to the condenser. The condenser cools the process stream to condense at least a portion of the elemental sulfur to produce a liquid sulfur stream (for example, the sulfur stream **129**). The sulfur stream **129** is separated from the process gas (syngas stream **121'**). For example, the condenser cools the process stream to a temperature that is cooler than the dew point of sulfur, such that at least a portion of the elemental sulfur in the process stream condenses and separates from a remaining gas portion of the process stream. In some implementations, the Claus unit **160** includes a multi-stage Claus process that includes multiple reactors (for example, a two-stage Claus process including two reactor trains) in series. In a two-stage Claus process, the process stream exiting the condenser of the first reactor train flows to the preheater of the second reactor train. In some implementations, the preheater of the second reactor train heats the process stream to provide a catalytic converter maximum bottom bed temperature that is about 10° C. hotter than the dewpoint of sulfur.

(13) The shift/hydrogenation reactor **130** is configured to receive the syngas stream **121'**. In some implementations, the system **100** includes a steam stream **125** that also flows to the shift/hydrogenation reactor **130**. Sufficient steam (from the steam stream **125**) is provided to the shift/hydrogenation reactor **130** to achieve equilibrium in the shift/hydrogenation reactor **130** for shifting carbon monoxide to carbon dioxide. Providing insufficient steam can retard the shift of carbon monoxide, and the shift reaction may not reach equilibrium, which is disadvantageous. In some implementations, the steam stream **125** mixes with the syngas stream **121'** before entering the shift/hydrogenation reactor **130**. In some implementations, the steam stream **125** and the syngas stream **121'** separately enter the shift/hydrogenation reactor **130** and mix within the shift/hydrogenation reactor **130**. The shift/hydrogenation reactor **130** includes a water-gas shift hydrogenation catalyst **132** that is configured to convert at least a portion of the carbon monoxide of the syngas stream **121'** and steam from the steam stream **125** to carbon dioxide and hydrogen to produce a shifted gas stream **131**. The residual portion of the sulfur dioxide in the syngas stream **121'** can be reduced back into hydrogen sulfide (reduction reaction:

$\text{SO}_2 + \text{H}_2 \rightarrow \text{S} + \text{H}_2\text{O}$) in the shift/hydrogenation reactor **130**. The sulfur (vapor) in the syngas stream **121'** can be reduced back into hydrogen sulfide (reduction reaction: $\text{S} + \text{H}_2 \rightarrow \text{H}_2\text{S}$) in the shift/hydrogenation reactor **130**. Other sulfur-containing compounds (such as carbonyl sulfide (COS) and carbon disulfide (CS_2)) in the syngas stream **121'** can be hydrolyzed to produce hydrogen sulfide. Because organic sulfur-containing compounds are hydrolyzed into hydrogen sulfide, downstream sulfur removal can be carried out more easily and more efficiently. Further, the sour shift reactions occurring in the shift/hydrogenation reactor

130 eliminates the need for separate COS hydrolysis conditioning. The shifted gas stream **131** exiting the shift/hydrogenation reactor **130** includes more carbon dioxide, more hydrogen, more hydrogen sulfide, and less carbon monoxide in comparison to the syngas stream **121'** entering the shift/hydrogenation reactor **130**. Because the water-gas shift reaction and reduction reactions are exothermic, the shifted gas stream **131** exiting the shift/hydrogenation reactor **130** is typically higher in temperature in comparison to the syngas stream **121'** entering the shift/hydrogenation reactor **130**. The steam stream **125** not only provides a source of water for facilitating the production of carbon dioxide and hydrogen, but also provides temperature regulation in the shift/hydrogenation reactor **130**.

(14) In some implementations, the shift/hydrogenation reactor **130** includes a pair of reactors with one of the reactors operating at a higher temperature in comparison to the other reactor. For example, a first shift reactor can operate at a temperature in a range of from about 350° C. to about 450° C., while the second shift reactor can operate at a temperature in a range of from about 180° C. to about 300° C. In order to control the operating temperature of the first shift reactor to be within a desired range, the flow rate of steam provided to the first shift reactor may be restricted due to the exothermic nature of the water-gas shift reactions. In some implementations, the first shift reactor includes a first water-gas shift hydrogenation catalyst, and the second shift reactor includes a second water-gas shift hydrogenation catalyst. In some implementations, the first and second water-gas shift hydrogenation catalysts are the same. In some implementations, the first and/or second water-gas shift hydrogenation catalysts are substantially similar to catalysts typically implemented for sulfur recovery unit (SRU) tail gas treatment processes. In some implementations, the first water-gas shift hydrogenation catalyst includes iron oxide, chromium oxide, magnesium oxide, cobalt-molybdenum, aluminum oxide (Al.sub.2O.sub.3), or any combination of these. In some implementations, the first water-gas shift hydrogenation catalyst is loaded on a ceramic support or an alumina support. In some implementations, the second water-gas shift hydrogenation catalyst includes copper, copper oxide, zinc oxide, aluminum oxide, cobalt-molybdenum, iron oxide, chromium oxide, magnesium oxide, or any combination of these. In some implementations, the second water-gas shift hydrogenation catalyst is loaded on a ceramic support or an alumina support. As one example, the first and second water-gas shift hydrogenation catalysts are cobalt-molybdenum based catalysts loaded on an alumina support. As another example, the first water-gas shift hydrogenation catalyst can be tailored to focus on hydrogenation reactions, while the second water-gas shift hydrogenation catalyst can be tailored to focus on water-gas shift reactions, or vice versa.

(15) In some implementations, the shift/hydrogenation reactor **130** is configured to convert at least 80%, at least 85%, at least 90%, or at least 95% of the carbon monoxide (CO) in the syngas stream **121'** into carbon dioxide (CO.sub.2). In some implementations, a molar ratio of hydrogen (H.sub.2) to carbon monoxide (CO) in the shifted gas stream **131** exiting the shift/hydrogenation reactor **130** is in a range of from 30:1 to 60:1. In some implementations, a molar ratio of hydrogen (H.sub.2) to carbon dioxide (CO.sub.2) in the shifted gas stream **131'** exiting the shift/hydrogenation reactor **130** is in a range of from 3:2 to 5:2. In some implementations, a molar ratio of hydrogen (H.sub.2) to water (H.sub.2O), for example, in the form of water vapor) in the shifted gas stream **131** exiting the shift/hydrogenation reactor **130** is in a range of from 3:2 to 5:2. In some implementations, a molar ratio of carbon dioxide (CO.sub.2) to water (H.sub.2O) in the shifted gas stream **131** exiting the shift/hydrogenation reactor **130** is in a range of from 1:2 to 3:2. For example, a molar ratio of hydrogen (H.sub.2) to carbon monoxide (CO) to carbon dioxide (CO.sub.2) to water (H.sub.2O) in the shifted gas stream **131** exiting the shift/hydrogenation reactor **130** is about 51:1:23:25. The residual portion of the sulfur dioxide and the sulfur (vapor) can be reduced back to hydrogen sulfide, such that no residuals remain in the shifted gas stream **131** exiting the shift/hydrogenation reactor **130**.

(16) The quench tower **140** is configured to receive the shifted gas stream **131** and separate water

(for example, bulk water phase) from the shifted gas stream **131** to produce a dehydrated gas stream **141**. In some implementations, the quench tower **140** receives a water stream **133**, and the quench tower **140** contacts the shifted gas stream **131** with the water stream **133**. Contacting the shifted gas stream **131** with the water stream **133** causes the shifted gas stream **131** to cool and also causes water in the shifted gas stream **131** (for example, in the form of water vapor) to condense and transfer to the water stream **133**. The dehydrated gas stream **141** and a reject water stream **143** discharge from the quench tower **140**. The dehydrated gas stream **141** exiting the quench tower **140** has a lower temperature in comparison to the shifted gas stream **131** entering the quench tower **140**. In some implementations, the dehydrated gas stream **141** exiting the quench tower **140** has an operating temperature in a range of from about 43° C. to about 60° C. The dehydrated gas stream **141** exiting the quench tower **140** has a lower water content in comparison to the shifted gas stream **131** entering the quench tower **140**. In some implementations, the dehydrated gas stream **141** is substantially free of water.

(17) The amine treater **170** includes an absorber **170a** and a regenerator **170b**. The absorber **170a** is configured to receive the dehydrated gas stream **141** and contact the dehydrated gas stream **141** with a lean amine solvent stream **171**. The lean amine solvent stream **171** has a high affinity for hydrogen sulfide, and in some cases, also for carbon dioxide. As the lean amine solvent stream **171** contacts the dehydrated gas stream **141**, the hydrogen sulfide separates from the dehydrated gas stream **141** and dissolves in the lean amine solvent stream **171**. That is, the lean amine solvent stream **171** extracts hydrogen sulfide from the dehydrated gas stream **141**. In some cases, the lean amine solvent stream **171** also extracts carbon dioxide from the dehydrated gas stream **141**. The dehydrated gas stream **141'** exiting the absorber **170a** is considered “sweet” and has a lower hydrogen sulfide content in comparison to the dehydrated gas stream **141** entering the absorber **170a**. In some implementations, the dehydrated gas stream **141'** exiting the absorber **170a** has a lower carbon dioxide content in comparison to the dehydrated gas stream **141** entering the absorber **170a**. The dehydrated gas stream **141'** exiting the absorber **170a** continues onto the purifier **150** for separating out the hydrogen product stream **151**.

(18) The lean amine solvent stream **171** that has absorbed the hydrogen sulfide (and, in some cases, carbon dioxide) is then considered a rich amine solvent stream **173**, which discharges from the absorber **170a**. The regenerator **170b** is configured to receive the rich amine solvent stream **173** and boil off the hydrogen sulfide (and, in some cases, carbon dioxide) from the rich amine solvent stream **173** to regenerate the lean amine solvent stream **171**. The regenerator **170b** can be, for example, a distillation column that fractionates the rich amine solvent stream **173** to produce the lean amine solvent stream **171** as a bottoms product and an acid gas stream **175** as a distillate. The lean amine solvent stream **171** is recycled to the absorber **170a**. The acid gas stream **175** includes the hydrogen sulfide that has been extracted from the dehydrated gas stream **141**. In some implementations, the acid gas stream **175** includes the carbon dioxide that has been extracted from the dehydrated gas stream **141**. At least a portion of (for example, all of) the acid gas stream **175** is recycled back to the combustion reactor **110**.

(19) The purifier **150** is configured to receive the dehydrated gas stream **141'** and separate hydrogen from the dehydrated gas stream **141'** to produce a hydrogen product stream **151** and an exhaust stream **153**. In some implementations, the system **100** includes a compressor **145** upstream of the purifier **150** to facilitate flow of the dehydrated gas stream **141'** to the purifier **150**. In some implementations, the compressor **145** is configured to pressurize the dehydrated gas stream **141'** to an operating pressure in a range of from about 1,000 kilopascals (kPa) to about 7,000 kPa. The hydrogen product stream **151** includes a majority of the hydrogen from the dehydrated gas stream **141'**. In some implementations, the hydrogen product stream **151** has a hydrogen content of at least 95 volume percent (vol. %), at least 96 vol. %, at least 97 vol. %, at least 98 vol. %, at least 99 vol. %, at least 99.5 vol. %, at least 99.9 vol. %, or at least 99.99 vol. %. The exhaust stream **153** includes a remainder of the dehydrated gas stream **141'**. For example, the exhaust stream **153** can

include carbon monoxide, carbon dioxide, and any hydrocarbon(s) that may have carried over through the system **100** without being combusted in the combustion reactor **110**. In some implementations, the system **100** produces about 0.1 kilograms (kg) to about 5 kg of carbon dioxide (CO.sub.2) per 1 kg of hydrogen (H.sub.2). That is, in some implementations, a mass ratio of carbon dioxide (CO.sub.2) in the exhaust stream **153** to hydrogen (H.sub.2) in the hydrogen product stream **151** is in a range of from about 0.1:1 to about 5:1. At least a portion of (for example, all of) the exhaust stream **153** can be recycled back to the combustion reactor **110**. By recycling at least the portion of the exhaust stream **153** back to the combustion reactor **110**, carbon dioxide that has been produced in the system **100** can be recycled within the system **100** without being released to the atmosphere and therefore avoid increasing emissions. In some implementations, a remaining portion of the exhaust stream **153** (not recycled to the combustion reactor **110**, shown by the dashed arrow branching from the exhaust stream **153**) can be sequestered, for example, within a subterranean formation, such that the carbon dioxide is not released to the atmosphere. In this way, the system **100** is a carbon neutral system, and carbon dioxide emissions by the system **100** can be avoided. In some implementations, the purifier **150** includes a membrane (for example, a polymer membrane) that selectively allows hydrogen to pass through (or pass more quickly in comparison to other components), such that the hydrogen can be separated from the remaining components of the dehydrated gas stream **141**. In some implementations, the purifier **150** includes a pair of vessels including an adsorbent, and the pair of vessels implement a pressure swing adsorption process to separate hydrogen from the remaining components of the dehydrated gas stream **141**.

(20) The system **100** can include a boiler to utilize heat generated in the system **100** to generate steam. In some implementations, as shown in FIG. 1, the system **100** includes a boiler **161**. Although shown in FIG. 1 as including one boiler (**161**), the system **100** can optionally include additional boilers (for example, two boilers, three boilers, or more than three boilers). The boiler **161** can generate steam **163** at varying operating pressures based on the quality of heat (for example, level of temperature) available in the system **100**. For example, the boiler **161** can be configured to use heat generated from the sub-stoichiometric combustion of the sour gas stream **101** within the combustion reactor **110** to generate steam. In using the heat from the combustion reactor **110**, the boiler **161** can cool the soot and sour syngas to a temperature in a range of from about 250° C. to about 500° C. In some implementations, as shown in FIG. 1, the boiler **161** is coupled to or at least partially reside within the combustion reactor **110** to receive heat from the combustion reactor **110**. In some implementations, at least a portion of the steam **163** generated by the boiler **161** is used to generate power. For example, at least a portion of the steam **163** generated by the boiler **161** can be flowed through a steam turbine **165**. The steam **163** flowing through the steam turbine **165** causes blades of the steam turbine **165** to rotate, and the steam turbine **165** generates electrical power in response to rotation of the blades of the steam turbine **165**. In some cases, at least a portion of the electrical power generated by the steam turbine **165** is used in the system **100**. For example, at least a portion of the electrical power generated by the steam turbine **165** can be used to operate the compressor **145** to pressurize the dehydrated gas stream **141'**. The remaining portion of the electrical power generated by the steam turbine **165** can be exported, for example, to a power grid or an end user. The steam **163** exiting the steam turbine **165** can be exported, for example, to a steam header for distribution to different users.

(21) In some implementations, a second boiler (not shown) can be configured to use heat from the soot and syngas to generate steam. In using the heat from the soot and syngas, the second boiler can cool the soot and syngas to a temperature in a range of from about 100° C. to about 200° C. In some implementations, the second boiler is located upstream of the mechanical separator **120**. In some implementations, a third boiler (not shown) can be configured to use heat from the shifted gas stream **131** to generate steam. In some implementations, the third boiler is located downstream of the shift/hydrogenation reactor **130** and upstream of the quench tower **140**. The steam stream **125**

can be at least partially sourced from any one or more of the first boiler **161**, second boiler, or third boiler, depending on the operating condition requirements for the steam stream **125**. For example, the steam stream **125** that is provided for the shift/hydrogenation reactor **130** can be at least partially sourced from the first boiler **161**, the second boiler, the third boiler, or any combination of these.

(22) In some implementations, as shown in FIG. **1**, the system **100** includes a feed pre-treatment unit **180** upstream of the combustion reactor **110**. The feed pre-treatment unit **180** can be configured to receive and process a high pressure raw sour gas stream **181** (for example, from a sour gas well and/or a gas-oil separation plant) to separate water **183** and natural gas liquids **185** from the high pressure raw sour gas stream **181**. For example, the water **183** and the natural gas liquids **185** can be condensed to separate from the sour gas stream **101**. The remaining portion of the high pressure raw sour gas stream **181** (excluding the water **183** and the natural gas liquids **185**) can be the sour gas stream **101**.

(23) In some implementations, as shown in FIG. **1**, the system **100** includes a turboexpander **187**. The sour gas stream **101** can flow through the turboexpander **187** to generate power. For example, at least a portion of (for example, all of) the sour gas stream **101** can be flowed through the turboexpander **187**. The sour gas stream **101** flowing through the turboexpander **187** causes blades of the turboexpander **187** to rotate, and the turboexpander **187** generates electrical power in response to rotation of the blades of the turboexpander **187**. In some cases, natural gas liquids **188** drop out (condense) from the sour gas stream **101** from flowing through the turboexpander **187**. The natural gas liquids **188** can be combined with the natural gas liquids **185**. In some cases, at least a portion of the electrical power generated by the turboexpander **187** is used in the system **100**. For example, at least a portion of the electrical power generated by the turboexpander **187** can be used to operate the compressor **145** to pressurize the dehydrated gas stream **141'**. The remaining portion of the electrical power generated by the turboexpander **187** can be exported, for example, to a power grid or an end user.

(24) Although not shown in FIG. **1**, the system **100** can (and is expected to) include the typical components included in similar systems. For example, in each of the configurations described, process streams (also referred to as “streams”) are flowed within each unit and between units of the respective system. The process streams can be flowed using one or more flow control systems implemented throughout the respective system. A flow control system can include one or more flow pumps to pump the liquid process streams, one or more blowers and/or compressors to flow the gaseous process streams, one or more flow pipes through which the process streams are flowed, and one or more flow elements (such as valves and orifice plates) to regulate the flow of streams through the pipes.

(25) In some implementations, a flow control system can be operated manually. For example, an operator can set a flow rate for each pump by changing the position of a valve (open, partially open, or closed) to regulate the flow of the process streams through the pipes in the flow control system. Once the operator has set the flow rates and the valve positions for all flow control systems distributed across the respective system, the flow control system can flow the streams within a unit or between units under constant flow conditions, for example, constant volumetric or mass flow rates. To change the flow conditions, the operator can manually operate the flow control system, for example, by changing the valve position.

(26) In some implementations, a flow control system can be operated automatically. For example, the flow control system can be connected to a computer system to operate the flow control system. The computer system can include a computer-readable medium storing instructions (such as flow control instructions) executable by one or more processors to perform operations (such as flow control operations). For example, an operator can set the flow rates by setting the valve positions for all flow control systems distributed across the respective system using the computer system. In such implementations, the operator can manually change the flow conditions by providing inputs

through the computer system. In such implementations, the computer system can automatically (that is, without manual intervention) control one or more of the flow control systems, for example, using feedback systems implemented in one or more units and connected to the computer system. For example, a sensor (such as a pressure sensor or temperature sensor) can be connected to a pipe through which a process stream flows. The sensor can monitor and provide operating conditions (such as a pressure or temperature) of the process stream to the computer system. In response to the operating condition deviating from a set point (such as a target pressure value or target temperature value) or exceeding a threshold (such as a threshold pressure value or threshold temperature value), the computer system can automatically perform operations to adjust properties of the flow control system. For example, if the pressure or temperature in the pipe exceeds the threshold pressure value or the threshold temperature value, respectively, the computer system can provide a signal to open a valve to relieve pressure or a signal to shut down process stream flow.

(27) FIG. 2 is a flowchart of an example method **200** for producing hydrogen from gasification of sour gas. The system **100** can, for example, implement the method **200**. At block **202**, a sour gas stream (such as the sour gas stream **101**) is combusted, in the presence of oxygen, within a reactor (such as the combustion reactor **110**) to produce soot (such as the soot **123**) and a sour syngas stream (such as the sour syngas stream **121**). Combusting the sour gas stream **101** within the reactor **110** at block **202** includes sub-stoichiometric combustion of the sour gas stream **101**. The sour gas stream **101** has a molar content of carbon (C) and a molar content of hydrogen (H). A molar content of oxygen gas in the reactor **110** at block **202** is in a range of from 10% to 70% of $C+H/4$. The sub-stoichiometric combustion of the sour gas stream **101** within the reactor **110** at block **202** converts at least 10% of the carbon in the sour gas stream **101** into the soot **123**. At block **204**, at least a portion of the hydrogen sulfide of the sour syngas stream **121** is reacted with sulfur dioxide (for example, from the sulfur dioxide stream **127**) to produce a syngas stream (such as the syngas stream **121'**). The Claus unit **160** can, for example, perform block **204**. The syngas stream **121'** includes the carbon dioxide from the sour syngas stream **121**, the carbon monoxide from the sour syngas stream **121**, the hydrogen from the sour syngas stream **121**, a residual portion of the hydrogen sulfide from the sour syngas stream **121** that did not react in the Claus unit **160**, a residual portion of sulfur dioxide that did not react in the Claus unit **160**, sulfur vapor, and water (for example, in the form of steam). Sulfur produced by the Claus unit **160** can be separated from the gas to produce the sulfur stream **129** (liquid). At block **206**, the syngas stream **121'** is contacted with a water-gas shift hydrogenation catalyst (such as the water-gas shift hydrogenation catalyst **132**) in a shift reactor (such as the shift/hydrogenation reactor **130**) in the presence of steam (such as steam from the steam stream **125**) to convert at least a portion of the carbon monoxide (from the syngas stream **121'**) and steam into carbon dioxide and hydrogen to produce a shifted gas stream (such as the shifted gas stream **131**). Contacting the syngas stream **121'** with the water-gas shift hydrogenation catalyst **132** in the shift/hydrogenation reactor **130** in the presence of steam at block **206** can also result in reducing the residual portion of sulfur dioxide and sulfur vapor into hydrogen sulfide, such that the shifted gas stream **131** is substantially free of sulfur dioxide and sulfur vapor. At block **208**, water is separated from the shifted gas stream **131** to produce a dehydrated gas stream (such as the dehydrated gas stream **141**). For example, the shifted gas stream **131** is contacted with water (such as the water stream **133**) in a quench tower (such as the quench tower **140**) at block **208**. Contacting the shifted gas stream **131** with the water stream **133** in the quench tower **140** at block **208** can cause the shifted gas stream **131** to cool and can also cause water in the shifted gas stream **131** (for example, in the form of water vapor) to condense and transfer to the water stream **133**. At block **210**, the dehydrated gas stream **141** is contacted with a lean amine solvent (such as the lean amine solvent stream **171**) to extract the residual portion of the hydrogen sulfide from the dehydrated gas stream **141** into the lean amine solvent stream **171** to produce a sweet gas stream (such as the dehydrated gas stream **141'**) and a rich amine solvent (such as the rich amine solvent stream **173**). The dehydrated gas stream **141'** produced at block **210** includes

carbon dioxide and hydrogen. The dehydrated gas stream **141'** produced at block **210** can also include hydrogen sulfide, carbon monoxide, and residual hydrocarbons. The absorber **170a** can, for example, perform block **210**. At block **212**, the residual portion of the hydrogen sulfide is boiled off from the rich amine solvent stream **173** to produce an acid gas stream (such as the acid gas stream **175**) and regenerate the lean amine solvent stream **171**. The acid gas stream **175** produced at block **212** includes the residual portion of the hydrogen sulfide, and in some cases, carbon dioxide. The regenerator **170b** can, for example, perform block **212**. In some implementations, at least a portion of (for example, all of) the acid gas stream **175** produced at block **212** is recycled to the combustion reactor **110**. At block **214**, the sweet gas stream (dehydrated gas stream **141'**) is separated to produce a hydrogen product stream (such as the hydrogen product stream **151**) and an exhaust stream (such as the exhaust stream **153**). The hydrogen product stream **151** produced at block **214** includes a majority of the hydrogen from the dehydrated gas stream **141'**. The exhaust stream **153** produced at block **214** includes a remainder of the dehydrated gas stream **141'** (for example, carbon dioxide). The purifier **150** can, for example, perform block **214**. In some implementations, at least a portion of the exhaust stream **153** produced at block **214** is recycled to the combustion reactor **110**. A remaining portion of the exhaust stream **153** produced at block **214** that is not recycled to the combustion reactor **110** can, for example, be sequestered within a subterranean formation (such as a disposal well) to avoid releasing carbon dioxide into the atmosphere.

(28) FIG. 3 is a flowchart of an example method **300** for producing hydrogen from gasification of sour gas. The system **100** can, for example, implement the method **300**. At block **302**, a sour gas stream (such as the sour gas stream **101**) is combusted, in the presence of oxygen, to produce soot (such as the soot **123**) and a sour syngas stream (such as the sour syngas stream **121**). Combusting the sour gas stream **101** at block **302** includes sub-stoichiometric combustion of the sour gas stream **101**. The sour gas stream **101** has a molar content of carbon (C) and a molar content of hydrogen (H). A molar content of oxygen gas at block **302** is in a range of from 10% to 70% of $C+H/4$. The sub-stoichiometric combustion of the sour gas stream **101** at block **302** converts at least 10% of the carbon in the sour gas stream **101** into the soot **123**. The combustion reactor **110** can, for example, perform block **302**. At block **304**, at least a portion of the hydrogen sulfide of the sour syngas stream **121** is reacted with sulfur dioxide (for example, from the sulfur dioxide stream **127**) to produce a syngas stream (such as the syngas stream **121'**). The Claus unit **160** can, for example, perform block **304**. The syngas stream **121'** includes the carbon dioxide from the sour syngas stream **121**, the carbon monoxide from the sour syngas stream **121**, the hydrogen from the sour syngas stream **121**, a residual portion of the hydrogen sulfide from the sour syngas stream **121** that did not react in the Claus unit **160**, and water (for example, in the form of steam). Sulfur produced by the Claus unit **160** can be separated from the gas, such that the syngas stream **121'** exiting the Claus unit **160** is free of elemental sulfur. At block **306**, at least a portion of the carbon monoxide of the syngas stream **121'** is reacted with steam (such as steam from the steam stream **125**) in the presence of a water-gas shift hydrogenation catalyst (such as the water-gas shift hydrogenation catalyst **132**) to produce a shifted gas stream (such as the shifted gas stream **131**). The shift/hydrogenation reactor **130** can, for example, perform block **306**. Reacting the carbon monoxide with steam at block **306** produces carbon dioxide and hydrogen. Thus, the shifted gas stream **131** produced at block **306** includes more carbon dioxide, more hydrogen, and less carbon monoxide in comparison to the syngas stream **121'**. At block **308**, water and the residual portion of the hydrogen sulfide are separated from the shifted gas stream **131** to produce a sweet gas stream (such as the dehydrated gas stream **141'**). The water can be separated from the shifted gas stream **131** at block **308**, for example, contacting the shifted gas stream **131** with water (such as the water stream **133**) in a quench tower (such as the quench tower **140**), causes the shifted gas stream **131** to cool and also causes water in the shifted gas stream **131** (for example, in the form of water vapor) to condense and transfer to the water stream **133**, thereby forming a dehydrated gas stream (such as the dehydrated gas stream **141**). Hydrogen sulfide can be separated from the shifted gas stream **131**

at block **308**, for example, by contacting a portion of the shifted gas stream **131** (such as the dehydrated gas stream **141**) with a lean amine solvent (such as the lean amine solvent stream **171**) to extract the residual portion of the hydrogen sulfide from the dehydrated gas stream **141** into the lean amine solvent stream **171** to produce the sweet gas stream (dehydrated gas stream **141'**) and a rich amine solvent (such as the rich amine solvent stream **173**). The dehydrated gas stream **141'** produced at block **308** includes carbon dioxide and hydrogen. The absorber **170a** can, for example, separate the hydrogen sulfide at block **308**. At block **310**, the sweet gas stream (dehydrated gas stream **141'**) is separated into a hydrogen product stream (such as the hydrogen product stream **151**) and an exhaust stream (such as the exhaust stream **153**). The hydrogen product stream **151** produced at block **310** includes a majority of the hydrogen from the dehydrated gas stream **141'**. The exhaust stream **153** produced at block **310** includes a remainder of the dehydrated gas stream **141'** (for example, carbon dioxide). The purifier **150** can, for example, perform block **310**. In some implementations, at least a portion of the exhaust stream **153** produced at block **310** is recycled to the combustion reactor **110**. A remaining portion of the exhaust stream **153** produced at block **310** that is not recycled to the combustion reactor **110** can, for example, be sequestered within a subterranean formation (such as a disposal well) to avoid releasing carbon dioxide into the atmosphere.

(29) While this specification contains many specific implementation details, these should not be construed as limitations on the scope of what may be claimed, but rather as descriptions of features that may be specific to particular implementations. Certain features that are described in this specification in the context of separate implementations can also be implemented, in combination, in a single implementation. Conversely, various features that are described in the context of a single implementation can also be implemented in multiple implementations, separately, or in any sub-combination. Moreover, although previously described features may be described as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can, in some cases, be excised from the combination, and the claimed combination may be directed to a sub-combination or variation of a sub-combination.

(30) As used in this disclosure, the terms “a,” “an,” or “the” are used to include one or more than one unless the context clearly dictates otherwise. The term “or” is used to refer to a nonexclusive “or” unless otherwise indicated. The statement “at least one of A and B” has the same meaning as “A, B, or A and B.” In addition, it is to be understood that the phraseology or terminology employed in this disclosure, and not otherwise defined, is for the purpose of description only and not of limitation. Any use of section headings is intended to aid reading of the document and is not to be interpreted as limiting; information that is relevant to a section heading may occur within or outside of that particular section.

(31) As used in this disclosure, the term “about” or “approximately” can allow for a degree of variability in a value or range, for example, within 10%, within 5%, or within 1% of a stated value or of a stated limit of a range.

(32) As used in this disclosure, the term “substantially” refers to a majority of, or mostly, as in at least about 50%, 60%, 70%, 80%, 90%, 95%, 96%, 97%, 98%, 99%, 99.5%, 99.9%, 99.99%, or at least about 99.999% or more.

(33) Values expressed in a range format should be interpreted in a flexible manner to include not only the numerical values explicitly recited as the limits of the range, but also to include all the individual numerical values or sub-ranges encompassed within that range as if each numerical value and sub-range is explicitly recited. For example, a range of “0.1% to about 5%” or “0.1% to 5%” should be interpreted to include about 0.1% to about 5%, as well as the individual values (for example, 1%, 2%, 3%, and 4%) and the sub-ranges (for example, 0.1% to 0.5%, 1.1% to 2.2%, 3.3% to 4.4%) within the indicated range. The statement “X to Y” has the same meaning as “about X to about Y,” unless indicated otherwise. Likewise, the statement “X, Y, or Z” has the same meaning as “about X, about Y, or about Z,” unless indicated otherwise.

(34) Particular implementations of the subject matter have been described. Other implementations, alterations, and permutations of the described implementations are within the scope of the following claims as will be apparent to those skilled in the art. While operations are depicted in the drawings or claims in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed (some operations may be considered optional), to achieve desirable results. In certain circumstances, multitasking or parallel processing (or a combination of multitasking and parallel processing) may be advantageous and performed as deemed appropriate.

(35) Moreover, the separation or integration of various system modules and components in the previously described implementations should not be understood as requiring such separation or integration in all implementations, and it should be understood that the described components and systems can generally be integrated together or packaged into multiple products.

(36) Accordingly, the previously described example implementations do not define or constrain the present disclosure. Other changes, substitutions, and alterations are also possible without departing from the spirit and scope of the present disclosure.

Embodiments

(37) In an example implementation (or aspect), a method comprises: combusting, in the presence of oxygen, a sour gas stream within a reactor to produce soot and a sour syngas stream comprising carbon dioxide, carbon monoxide, hydrogen, and hydrogen sulfide, wherein the sour gas stream comprises hydrogen sulfide and at least one hydrocarbon, wherein combusting the sour gas stream within the reactor comprises sub-stoichiometric combustion of the sour gas stream, wherein the sour gas stream has a molar content of carbon (C) and a molar content of hydrogen (H), wherein a molar content of oxygen gas in the reactor is in a range of from 10% to 70% of $C+H/4$, wherein the sub-stoichiometric combustion of the sour gas stream within the reactor converts at least 10% of the carbon in the sour gas stream into the soot; reacting at least a portion of the hydrogen sulfide of the sour syngas stream with sulfur dioxide to produce a syngas stream comprising the carbon dioxide, the carbon monoxide, the hydrogen, water, elemental sulfur vapor, a residual portion of the hydrogen sulfide, and a residual portion of the sulfur dioxide, wherein reacting at least the portion of the hydrogen sulfide of the sour syngas stream with sulfur dioxide produces a sulfur stream comprising elemental sulfur liquid; contacting, in the presence of steam, the syngas stream with a water-gas shift hydrogenation catalyst in a shift reactor to convert at least a portion of the carbon monoxide and steam to carbon dioxide and hydrogen to produce a shifted sour gas stream, wherein contacting the syngas stream with the water-gas shift hydrogenation catalyst in the shift reactor in the presence of steam reduces the elemental sulfur vapor and the residual portion of the sulfur dioxide into hydrogen sulfide, and the shifted sour gas stream is substantially free of elemental sulfur and sulfur dioxide; separating water from the shifted sour gas stream to produce a dehydrated sour gas stream; contacting the dehydrated sour gas stream with a lean amine solvent to extract the residual portion of the hydrogen sulfide from the dehydrated sour gas stream into the lean amine solvent to produce a sweet gas stream and a rich amine solvent, the sweet gas stream comprising the carbon dioxide and the hydrogen; boiling off the residual portion of the hydrogen sulfide from the rich amine solvent to produce an acid gas stream and regenerate the lean amine solvent, the acid gas stream comprising the residual portion of the hydrogen sulfide; and separating the sweet gas stream to produce a hydrogen product stream and an exhaust stream, the hydrogen product stream comprising a majority of the hydrogen from the sweet gas stream, and the exhaust stream comprising a remainder of the sweet gas stream.

(38) In an example implementation (or aspect) combinable with any other example implementation (or aspect), at least a portion of heat generated from sub-stoichiometric combustion of the sour gas stream within the reactor is directed to a boiler to generate steam and cool the soot and sour syngas to a temperature in a range of from about 250 degrees Celsius ($^{\circ}\text{C.}$) to about 500 $^{\circ}\text{C.}$

(39) In an example implementation (or aspect) combinable with any other example implementation

(or aspect), at least a portion of the steam stream is sourced from the steam generated by the boiler.

(40) In an example implementation (or aspect) combinable with any other example implementation (or aspect), at least a portion of the steam generated by the boiler is flowed through a steam turbine, wherein the steam turbine generates electrical power in response to flow of the portion of the steam through the turbine.

(41) In an example implementation (or aspect) combinable with any other example implementation (or aspect), separating water from the shifted sour gas stream comprises contacting the shifted sour gas stream with a water stream within a quench tower and discharging a reject water stream and the dehydrated sour gas stream from the quench tower.

(42) In an example implementation (or aspect) combinable with any other example implementation (or aspect), at least a portion of the acid gas stream is recycled to the reactor.

(43) In an example implementation (or aspect) combinable with any other example implementation (or aspect), at least a portion of the exhaust stream is recycled to the reactor.

(44) In an example implementation (or aspect) combinable with any other example implementation (or aspect), a second portion of the exhaust stream is sequestered within a subterranean formation to avoid releasing carbon dioxide to the atmosphere.

(45) In an example implementation (or aspect) combinable with any other example implementation (or aspect), a high pressure raw sour gas stream is processed in a feed pre-treatment unit, wherein processing the high pressure raw sour gas stream in the feed pre-treatment unit comprises: separating water from the high pressure raw sour gas stream; and condensing natural gas liquids from the high pressure raw sour gas stream to produce a natural gas liquids (NGL) stream and the sour gas stream.

(46) In an example implementation (or aspect) combinable with any other example implementation (or aspect), the sour gas stream is flowed through a turboexpander prior to combusting the sour gas stream within the reactor, wherein the turboexpander generates electrical power in response to flow of the sour gas stream through the turboexpander.

(47) In an example implementation (or aspect), a system comprises: a sour gas stream comprising hydrogen sulfide and at least one hydrocarbon, the sour gas stream having a molar flow rate of carbon (C) and a molar flow rate of hydrogen (H); an oxidizing stream comprising oxygen gas and having a molar flow rate of oxygen gas (O_{2}) in a range of from 10% to 70% of $C+H/4$; a combustion reactor configured to receive the sour gas stream and the oxidizing stream, the combustion reactor configured to combust the sour gas stream to produce soot and sour syngas, wherein sub-stoichiometric combustion of the sour gas stream within the combustion reactor converts at least 10% of the carbon in the sour gas stream into soot; a mechanical separator configured to separate the soot from the sour syngas to produce a sour syngas stream comprising carbon dioxide, carbon monoxide, hydrogen, and hydrogen sulfide; a Claus unit configured to receive the sour syngas stream and a sulfur dioxide stream comprising sulfur dioxide, the Claus unit configured to react at least a portion of the hydrogen sulfide of the sour syngas stream with the sulfur dioxide of the sulfur dioxide stream to produce elemental sulfur and water, the Claus unit configured to discharge a sulfur stream comprising the elemental sulfur, the Claus unit configured to discharge a syngas stream comprising the carbon dioxide, the carbon monoxide, the hydrogen, the water, elemental sulfur vapor, a residual portion of the hydrogen sulfide, and a residual portion of the sulfur dioxide; a sour shift/hydrogenation reactor configured to receive the syngas stream and steam, the sour shift/hydrogenation reactor comprising a water-gas shift hydrogenation catalyst, the water-gas shift hydrogenation catalyst configured to convert at least a portion of the carbon monoxide of the syngas stream and steam to carbon dioxide and hydrogen to produce a shifted sour gas stream, the water-gas shift hydrogenation catalyst configured to reduce the elemental sulfur vapor and the residual portion of the sulfur dioxide into hydrogen sulfide, the shifted sour gas stream substantially free of elemental sulfur and sulfur dioxide; a quench tower configured to receive the shifted sour gas stream and separate water from the shifted sour gas

stream to produce a dehydrated sour gas stream; an amine unit configured to receive the dehydrated sour gas stream, the amine unit configured to contact the dehydrated sour gas stream with a lean amine solvent to extract the residual portion of the hydrogen sulfide from the dehydrated sour gas stream into the lean amine solvent to produce a sweet gas stream and a rich amine solvent, the sweet gas stream comprising the carbon dioxide and the hydrogen, the amine unit configured to boil off the residual portion of the hydrogen sulfide from the rich amine solvent to produce an acid gas stream and regenerate the lean amine solvent, the acid gas stream comprising the residual portion of the hydrogen sulfide; and a purifier configured to receive the sweet gas stream and separate hydrogen from the sweet gas stream to produce a hydrogen product stream and an exhaust stream, the hydrogen product stream comprising a majority of the hydrogen from the sweet gas stream, the exhaust stream comprising a remainder of the sweet gas stream, wherein the combustion reactor is configured to receive at least a portion of the exhaust stream.

(48) In an example implementation (or aspect) combinable with any other example implementation (or aspect), the system comprises a boiler configured to use heat generated from combustion of the sour gas stream within the combustion reactor to generate steam and cool the soot and sour syngas to a temperature in a range of from about 250 degrees Celsius ($^{\circ}$ C.) to about 500 $^{\circ}$ C.

(49) In an example implementation (or aspect) combinable with any other example implementation (or aspect), the sour shift/hydrogenation reactor is configured to receive at least a portion of the steam generated by the boiler.

(50) In an example implementation (or aspect) combinable with any other example implementation (or aspect), the system comprises a steam turbine configured to receive at least a portion of the steam generated by the boiler, the steam turbine configured to generate electrical power in response to the portion of the steam flowing through the steam turbine.

(51) In an example implementation (or aspect) combinable with any other example implementation (or aspect), the combustion reactor is configured to receive at least a portion of the acid gas stream.

(52) In an example implementation (or aspect) combinable with any other example implementation (or aspect), a second portion of the exhaust stream is sequestered within a subterranean formation to avoid releasing carbon dioxide to the atmosphere.

(53) In an example implementation (or aspect) combinable with any other example implementation (or aspect), the system comprises a feed pre-treatment unit upstream of the combustion reactor, wherein the feed pre-treatment unit is configured to process a high pressure raw sour gas stream to separate water and natural gas liquids from the high pressure raw sour gas stream to produce a reject water stream, a natural gas liquids (NGL) stream, and the sour gas stream.

(54) In an example implementation (or aspect) combinable with any other example implementation (or aspect), the system comprises a turboexpander downstream of the feed pre-treatment unit and upstream of the combustion reactor, wherein the turboexpander is configured to receive the sour gas stream and generate electrical power in response to flow of the sour gas stream through the turboexpander, wherein the turboexpander is configured to discharge the sour gas stream to the combustion reactor.

(55) In an example implementation (or aspect), a method comprises: combusting, in the presence of oxygen, a sour gas stream to produce soot and a sour syngas stream comprising carbon dioxide, carbon monoxide, hydrogen, and hydrogen sulfide, wherein the sour gas stream comprises hydrogen sulfide and at least one hydrocarbon, wherein combusting the sour gas stream comprises sub-stoichiometric combustion of the sour gas stream, wherein the sour gas stream has a molar content of carbon (C) and a molar content of hydrogen (H), wherein a molar content of the oxygen that is present is in a range of from 10% to 70% of $C+H/4$, wherein the sub-stoichiometric combustion of the sour gas stream converts at least 10% of the carbon in the sour gas stream into the soot; reacting at least a portion of the hydrogen sulfide of the sour syngas stream with sulfur dioxide to produce a syngas stream comprising the carbon dioxide, the carbon monoxide, the hydrogen, water, elemental sulfur vapor, a residual portion of the hydrogen sulfide, and a residual

portion of the sulfur dioxide, wherein reacting at least the portion of the hydrogen sulfide of the sour syngas stream with sulfur dioxide produces a sulfur stream comprising elemental sulfur liquid; reacting, in the presence of a water-gas shift hydrogenation catalyst, at least a portion of the carbon monoxide of the syngas stream with steam to produce a shifted sour gas stream, wherein the shifted gas sour stream comprises more carbon dioxide, more hydrogen, more hydrogen sulfide, and less carbon monoxide in comparison to the syngas stream prior to reacting the carbon monoxide, and the shifted sour gas stream is substantially free of elemental sulfur and sulfur dioxide; separating water and the residual portion of the hydrogen sulfide from the shifted sour gas stream to produce a sweet gas stream; and separating the sweet gas stream into a hydrogen product stream and an exhaust stream, wherein the hydrogen product stream comprises a majority of the hydrogen from the sweet gas stream, and the exhaust stream comprises a remainder of the sweet gas stream.

(56) In an example implementation (or aspect) combinable with any other example implementation (or aspect), at least a portion of heat generated from combustion of the sour gas stream is directed to a boiler to generate steam and using at least a portion of the generated steam to react with at least the portion of the carbon monoxide of the syngas stream in the presence of the water-gas shift hydrogenation catalyst.

Claims

1. A system comprising: a sour gas stream comprising hydrogen sulfide and at least one hydrocarbon, the sour gas stream having a molar flow rate of carbon (C) and a molar flow rate of hydrogen (H); an oxidizing stream comprising oxygen gas and having a molar flow rate of oxygen gas (O_{2}) in a range of from 10% to 70% of $C+H/4$; a combustion reactor configured to receive the sour gas stream and the oxidizing stream, the combustion reactor configured to combust the sour gas stream to produce soot and sour syngas, wherein sub-stoichiometric combustion of the sour gas stream within the combustion reactor converts at least 10% of the carbon in the sour gas stream into soot; a mechanical separator configured to separate the soot from the sour syngas to produce a sour syngas stream comprising carbon dioxide, carbon monoxide, hydrogen, and hydrogen sulfide; a Claus unit configured to receive the sour syngas stream and a sulfur dioxide stream comprising sulfur dioxide, the Claus unit configured to react at least a portion of the hydrogen sulfide of the sour syngas stream with the sulfur dioxide of the sulfur dioxide stream to produce elemental sulfur and water, the Claus unit configured to discharge a sulfur stream comprising the elemental sulfur, the Claus unit configured to discharge a syngas stream comprising the carbon dioxide, the carbon monoxide, the hydrogen, the water, elemental sulfur vapor, a residual portion of the hydrogen sulfide, and a residual portion of the sulfur dioxide; a sour shift/hydrogenation reactor configured to receive the syngas stream and steam, the sour shift/hydrogenation reactor comprising a water-gas shift hydrogenation catalyst, the water-gas shift hydrogenation catalyst configured to convert at least a portion of the carbon monoxide of the syngas stream and steam to carbon dioxide and hydrogen to produce a shifted sour gas stream, the water-gas shift hydrogenation catalyst configured to reduce the elemental sulfur vapor and the residual portion of the sulfur dioxide into hydrogen sulfide, the shifted sour gas stream substantially free of elemental sulfur and sulfur dioxide; a quench tower configured to receive the shifted sour gas stream and separate water from the shifted sour gas stream to produce a dehydrated sour gas stream; an amine unit configured to receive the dehydrated sour gas stream, the amine unit configured to contact the dehydrated sour gas stream with a lean amine solvent to extract the residual portion of the hydrogen sulfide from the dehydrated sour gas stream into the lean amine solvent to produce a sweet gas stream and a rich amine solvent, the sweet gas stream comprising the carbon dioxide and the hydrogen, the amine unit configured to boil off the residual portion of the hydrogen sulfide from the rich amine solvent to produce an acid gas stream and regenerate the lean amine solvent, the acid gas stream comprising the residual portion of the hydrogen sulfide; and

- a purifier configured to receive the sweet gas stream and separate hydrogen from the sweet gas stream to produce a hydrogen product stream and an exhaust stream, the hydrogen product stream comprising a majority of the hydrogen from the sweet gas stream, the exhaust stream comprising a remainder of the sweet gas stream, wherein the combustion reactor is configured to receive at least a portion of the exhaust stream.
2. The system of claim 1, comprising a boiler configured to use heat generated from combustion of the sour gas stream within the combustion reactor to generate steam and cool the soot and sour syngas to a temperature in a range of from about 250 degrees Celsius ($^{\circ}$ C.) to about 500 $^{\circ}$ C.
 3. The system of claim 2, wherein the sour shift hydrogenation reactor is configured to receive at least a portion of the steam generated by the boiler.
 4. The system of claim 2, comprising a steam turbine configured to receive at least a portion of the steam generated by the boiler, the steam turbine configured to generate electrical power in response to the portion of the steam flowing through the steam turbine.
 5. The system of claim 1, wherein the combustion reactor is configured to receive at least a portion of the acid gas stream.
 6. The system of claim 1, wherein a second portion of the exhaust stream is sequestered within a subterranean formation to avoid releasing carbon dioxide to the atmosphere.
 7. The system of claim 1, comprising a feed pre-treatment unit upstream of the combustion reactor, wherein the feed pre-treatment unit is configured to process a high pressure raw sour gas stream to separate water and natural gas liquids from the high pressure raw sour gas stream to produce a reject water stream, a natural gas liquids (NGL) stream, and the sour gas stream.
 8. The system of claim 7, comprising a turboexpander downstream of the feed pre-treatment unit and upstream of the combustion reactor, wherein the turboexpander is configured to receive the sour gas stream and generate electrical power in response to flow of the sour gas stream through the turboexpander, wherein the turboexpander is configured to discharge the sour gas stream to the combustion reactor.
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